Divyansh Singh
X - A

II PERIODIC TEST : 2025-26

Class - X

Subject - Artificial Intelligence (417)

Time : 2 hrs. M.M.: 50

General Instruction :

1. Please read the instructions carefully.
2. This question paper consists of two sections: Section A and Section B
3. Section A has Objective type questions (MCQ, True/False and Fill in the blanks), whereas Section B contains Subjective type questions.
4. All questions of a particular section must be attempted in correct order.
5. Marks allotted are mentioned against each question/part.

SECTION A : Objective Type Questions

Q.1- *Multiple choice questions:-* [10]

i. Which of the following is a limitation of rule-based AI model?
   a. They can update their rules automatically.
   b. They adapt to new data easily.
   c. They do not modify their rules after training.
   d. They do not require predefined rules.

ii. What role do neural networks play in Deep Learning?
   a. They create static algorithms.
   b. They process large amount of data and recognize patterns.
   c. They identify anomalies in simple datasets.
   d. They only function for rule-based systems.

iii. What is the purpose of online evaluation?
   a. To test a model before deployment.
   b. To track model performance in real-time.
   c. To delete unnecessary data from the dataset.
   d. To track model on real-world data.

iv. Which factor can cause AI models to make incorrect predictions?
   a. High-quality training data.
   b. Proper error analysis.
   c. Poor-quality data.
   d. 100% accuracy.

v. What is the primary role of data science in decision-making?
   a. To predict future outcomes.
   b. To summarise past data.
   c. To make informed, evidence-based decisions.
   d. To analyse any visual data.

vi. What is one major limitation of No-Code AI tools?
   a. They require extensive coding knowledge.
   b. They lack flexibility for customization.
   c. They are only used in gaming applications.
   d. They are extremely expensive.

vii. Which of the following is an example of a No-Code AI tools?
   a. Python
   b. JavaScript
   c. Orange Data Mining
   d. TensorFlow

viii. Which of the following statements is true about Jupyter Notebook?
   a. Anaconda is web-based tool for installing Jupyter Notebook.
   b. Anaconda is a command-line interface tool for installing Jupyter Notebook.
   c. Anaconda can only be used from a web browser to install Jupyter Notebook.
   d. Anaconda is used to write Jupyter Notebooks, not install them.

Contd.....2-

ix. How do you access the last character of a string using negative indexing?
   a. String[-0] b. String[-1]
   c. String[-2] d. String[-3]

x. What is the main advantage of using NumPy arrays over Python lists for numerical computations?
   a. They can store mixed data types.
   b. They support direct arithmetic operations.
   c. They consume more money.
   d. They are slower for computations.

Q.2- *Fill in the blanks:-* *[10 × 1 = 10]*

i. AI systems rely on large amounts of _____ to make accurate decisions.
ii. The process where a neural network adjusts its weights based on errors is called _____ .
iii. In Deep Learning, _____ are used to process information and recognize patterns.
iv. The proportion of correct predictions made by an AI model is called _____ .
v. _____ tuning helps improve a model's accuracy by optimizing its setting like learning rate and number of layers.
vi. _____ occurs when model memorises patterns from training data but fails to generalize to new data.
vii. Machine learning is a subset of artificial intelligence that focuses on developing algorithms to enable machines to make _____ or decision based data.
viii. Data science involves collecting, analyzing, and using data to solve problems or make _____ decisions.
ix. The _____ attribute of sort( ) method is used to reverse the order of the list.
x. NumPy arrays are _____ because they can only store one type of data.

Q.3- *State True False:-* *[10 × 1= 10]*

i. A testing dataset is used to train the AI models.
ii. The output layer of a neural network processes data before presenting it to the user.
iii. Error analysis is not necessary for improving AI models.
iv. AI can only be used in computers and not in mobile devices.
v. AI helps machines make decisions based on data.
vi. Machine learning is a part of data science that helps analyse past data to inform future decisions.
vii. Teachable Machine allows users to create machine learning models without coding.
viii. The mean and median are always same in a normal distribution.
ix. You can access tuple entry using indexing.
x. Lists cannot store different types of elements.

SECTION B : Subjective Type Questions

Short Answer Questions:-

Q.4- Why is backward propagation important in a neural network? [2]
Q.5- What is the significance of errors in AI model? [2]
Q.6- What are two advantages using No-Code AI? [2]
Q.7- What is the purpose of the np.linspace( ) function? [2]

Long Answer Questions:-

Q.8- What factors contribute to errors in AI models, and how can they be minimized? [3]
Q.9- Describe the types of data distributions and provide examples for each. [3]

Python Programs:-

Q.10- Create a DataFrame that contains information about three individuals, including their names, ages, and cities of residence. The DataFrame should have the following columns :
   Name : Rahul, Amit, and Sandeep
   Age : 38, 40, and 42
   City : Sitapur, Hardoi and Kanpur [3]

Q.11- Write a program to create a bar graph showing the number of fruit sold. The data is as follows :
   Mangos : 10
   Apples : 5
   Oranges : 11 [3]

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Divyansh Singh

II PERIODIC TEST: 2025-26
CLASS X (CBSE)

TIME : 2 hrs. **SUBJECT - SOCIAL SCIENCE** **M.M. :50**

General Instructions:-

(i) The Question paper contains 23 Questions.
(ii) The question paper has Four Sections- A- History, B- Geography C- Political Science, and D- Economics.
(iii) Each Section is of 13 or 12 marks and has MCQS, VSA, SA, LAs and CBQ.
(iv) Very short answer type questions (VSA), carry 2 marks each. Answers to each question should not exceed 40 words.
(v) Short Answer Type Questions (SA), carry 3 marks each. Answers to each question should not exceed 60 words.
(vi) Long answer type Questions (LA), carry 5 marks each. Answer to each question should not exceed 120 words.
(vii) There are case based questions (CBQ) with three sub questions and are of 4 marks each. Answer to each question should not exceed 100 words.
(viii) The map- based questions, carry 5 marks with two parts- Q6 . In Section A- History (2 marks) and Q.11 in Section B- Geography (3 marks).
(ix) There is no overall choice in the question paper. However, an internal choice has been provided in few questions. Only one of the choice in such questions must be attempted.
(xi) In addition to this, separate instructions are given with each section and question, wherever necessary.

SECTION- 'A' HISTORY (13 MARKS)

Q.1 Which region was not included within the Habsburg Empire? [1]
a. France b. Tyrol
c. Bohemia d. Lombardy

Q.2 Arrange the following events of Indian National Movement in chronological order and choose the correct option. [1]
i. Formation of Swaraj Party
ii. Lahore session of the Indian National Congress.
iii. Gandhi- Irwin Pact
iv. Formation of Depressed Class Association.
a. ii, iii, i, iv b. i, ii, iv, iii
c. i, iii, iv, ii d. iv, iii, ii, i

Q.3 Who was proclaimed as king of United Italy in 1861? [2]

Q.4 Explain the Napoleonic Code. [2]

Q.5 Explain the limitations of the 'Civil Disobedience Movement'. [5]

OR

Explain the implications of the 'First World Ware' on the economic and political situation of India.

Q.6 Map Based Questions [2]
Two places A and B have been marked on the given outline map of India. Identify them (A) the place where Jallian walal Bagh Masses took place. (B) The place where Gandhiji Lake the Salt Law.

'SECTION- 'B' GEOGRAPHY (13 MARKS)

Q.7 Shiva gives his friend three clues about a type of soil. [1]
i. It is found in Western Ghats region of Maharashtra.
ii. Acidic in nature.
iii. They are prone to erosion and degradation. What soil is being referred to by shiva?
a. Black soil b. Arid soil
c. Red soil d. laterite soil

Q.8 Which of the following best defines plantation farming? [1]

a. Farming practiced on small family- owned plaots.
b. Growing multiple crops on a small piece of land.
c. Cultivating a single crop over extensive land area.
d. Growing crops using organic methods only.

Q.9 Which are the two types of iron- ore found in India? Write two features of each. [3]

Q.10 Explain any five technological and institutional reforms in the Indian agriculture. [5]

OR

Why is agriculture called the backbone of Indian economy? Explain.

Q.11. On the same outline map of India. Locate and label any three of the following: [3]

i. Hirakund Dam
ii. Tarapur Nuclear Power Plant
iii. Bokaro Coal Mines
iv. Largest surgarcane producing state.

'SECTION- 'C' POLITICAL SCIENCE (12 MARKS)

Q.12 What significant political changes occurred in India after 1990? [1]

i. Emergence of regional political parties.
ii. Coalition governments at the Center.
iii. Formation of alliances between national and regional parties.
iv. Single- party dominance in the Lok Sabha.

a. I and II
b. II and III
c. I, II and III
d. I and III

Q.13 Which one of the following statements best describes women empowerment? [1]

a. Encourage men to take leadership roles.
b. Promoting superiority of women over men.
c. Ensuring equal opportunities and rights for women.
d. Limiting the rights of men in all spheres.

Q.14 What is Majoritarianism. [2]

Q.15 Mention the provisions that constitute India into a secular country. [2]

Q.16 Why is power sharing desirable? [2]

Q.17 Read the given extract and answer all questions that follow.

The exact balance of power between the central and the state government varies from one federation to another. This balance depends mainly on the historical context in which the federation was formed. There are two kinds of routes through which federations have been formed. The first route involves independent states coming together on their own to form a bigger unit, so that by pooling sovereignty and retaining identity they can increase their security. This type of 'coming together federation' include the USA, Switzerland and Australia. The second route is where a large country decides to divide its power between the constituent states and the national government India, Spain and Belgium are the examples of this kind of Holding together Federation.'

17.1 On what does the balance of power between the central and state government depends. [1]

17.2 What are the two types of federations? [1]

17.3 Define 'sovereignty'. [2]

'SECTION- 'D' ECONOMICS (12 MARKS)

Q.18 Literacy rate measures the proportion of literate population in the _____ and above age group. [1]

a. five
b. six
c. seven
d. eight

Q.19 ~~Ibrahim gives a cheque of Rs.10,000 to Sahil. What type of medium of exchange did he use?~~ Life Insurance is an activity of [1]

a. ~~Paper note~~ Primary Sector
b. ~~Metallic coins~~ Secondary Sector
c. ~~Barter system~~ Servia Sector
d. ~~Demand deposit~~ None of these

Q.20 Explain the meaning of globalization? [1]

Q.21 ~~Why is cheap and affordable credit important for the country's development? Explain.~~ What is per capita income? [2]

Q.22 Explain three reasons for the rising importance of tertiary sector in India? [3]

Q.23 Read the following source and answer the questions that follow:

The liberalisation of foreign trade and investment in India was supported by some very powerful international organisations. These orgainisations say that all barriers to foreign trade and investment are harmful. There should be no barriers. World Trade Organisation (WTO) is one such organisation whose aim is to liberalise international trade. Started as the initiative of the developed countries. WTO establishes rules regarding international trade and sees that these rules are obeyed. As on July 2016, nearly 165 Countries of the World are currently member of the WTO.

Though WTO is supposed to allow free trade for all, in practice, it is seen that the developed countries have unfairly retained trade barriers. On the other hand, WTO rules have forced the developing countries to remove trade barriers.

23.1 How many countries are the member of World Trade Organisation as per July' 2016. [1]

23.2 Is there free trade in practice? [1]

23.3 Name one international trade organisation? Give its main aim. [2]

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II PERIODIC TEST- 2025-26

Time : 2 hrs. **CLASS - X (CBSE)** **M.M.: 50**

SUBJECT- ENGLISH LANGUAGE AND LITERATURE (184)

GENERAL INSTRUCTIONS-:

i. The Question Paper Contains three sections:
- Section- A Reading Skills
- Section- B Writing and Grammar
- Section- C Literature and Supplementary

ii Attempt questions based on specific instructions for each part.

SECTION- A : (READING SKILLS)

Q.1 Read the passage given below:

"Who doesn't know to cook rice? Cooking rice hardly takes time," said my father. So, I challenged myself. I switched from news to You Tube and typed, "How to cook rice?" I took one and a half cups of rice. Since I didn't have access to a rice cooker, I put the rice in a big pot. Firstly, the rice has to be washed to get rid of dust and starch. I thought I won't be able to drain the rice and that it will fall out of the pot. I observed the chef as I swirled the rice around and used my dexterous hands to drain it, not once, not twice, but three times. I looked down at the sink and saw less than 50 grains that made their way out of the pot. Suffice to say, I was up to the mark.

The video stated that the key to perfect rice is equal amount of rice and water. I have heard that professionals don't need to measure everything; they just know what the right amount is. But as this was my first time in the kitchen, I decided to experiment by not measuring the water needed for boiling the rice. I wanted the rice to be firm when bitten, just like pasta. I don't enjoy the texture of mushy rice. It has to have that chutzpah; it has to resist my biting power just for a bit before disintegrating.

After what seemed like 10 minutes, all the water disappeared. I went in to give it a good stir. To my surprise, some of the rice got stuck to the pot. I tried to scrape it off but to no avail. At the same time, there was a burning smell coming from it. I quickly turned the stove off.

The experience was fairly rewarding and memorable. It taught me a new sense of respect for those who cook food on a regular basis at home or are engaged in gourmet creations professionally.

Answer the following questions, based on the passage above. (ANY SIX) [6]

(i) Father's question to the narrator, about knowing how to cook rice, was intended to:
- a) criticise the narrator's lack of abilities.
- b) make the process sound simple.
- c) encourage the narrator to take up cooking.
- d) showcase his own expertise in cooking rice.

(ii) How did the narrator describe the desired texture of the cooked rice?

(iii) Complete the sentence appropriately.
After observing the rice draining process, the narrator felt confident because _____

(iv) In the given passage, the word "dexterous" is used to describe the hands used to drain the rice. Which word can be a synonym for "dexterous"?

(v) While cooking rice for the first time, the author encountered a challenging situation when some of the rice got stuck to the pot. They tried to scrape it off but failed and a _____ smell filled the kitchen.

(vi) Based on the reading of the passage, what lesson did the author learn from their experience of cooking rice for the first time? Explain.

(vii) Which option represents the correct ratio of water to rice for cooking perfect rice?

(1) 

(2) 

(3) 

(4) 

A. Image 1 B. Image 2 C. Image 3 D. Image 4

Q.2 Read the following passage carefully-:

A day after Argentina won the World Cup, Messi's picture of kissing the trophy became the most liked post on Instagram, surpassing likes on an egg! Then, a campaign to beat the egg started when a journalist noticed that Messi's photo was only 11 million likes away from being number one. "If you liked this egg, unlike it," he tweeted jokingly. But since it was after the Cup, people went berserk.

Fans started to encourage friends and followers to unlike the "World Record Egg," a parody account. It lost 150,000 likes and Messi gained 6 million by evening. Twelve hours later, Messi stood at more than 74 million likes. The fans broke the records at such short notice with passion and eccentricity.

A marketing agency created a website tracking the likes and allowing the users to subtract their likes from the egg and then add one to Messi's post. It peaked at 6,000 users in real time and had over 50,000 visitors during that week.

The fact that the campaign was centered on Instagram was crucial. On Twitter, powerful bots can impact the turn out. On Instagram, at least in this case, there is no reason to believe any artificial users took part in the process. It was 100% the fans.

Analysts believe that what led supporters to engage in the campaign was linked to a deep-seated desire to be part of the triumph they'd seen on the football pitch and wished to contribute to this victory.

Attempt the following questions on the basis of the passage you have read:

(a) Complete the following appropriately, based on your understanding of paragraphs 1 and 2. We can say that powerful bots impacting the turnout of popular and viral campaigns on Twitter is comparable with scripted reality shows on TV but the success of Messi's campaign on Instagram was driven .....
[1]

(b) Fill in the blanks with the appropriate option from those given in brackets, based on you understanding of paragraphs 3. [1]
A marketing agency jumped into the melee to keep a track of likes both on the egg and Messi's picture and analyzed the web traffic in real time is a ..... (fact/ option) because it is a/an ..... (subjective judgement/ objective detail)

(c) Justify the following in about 40 words. [2]
Such is the power of social media that even trivial comments can go viral on these platforms.

(d) (A) Assertion- Analysts believe supports engage in campaign to high extend. [1]
(R) Reason- Supporters are driven by aspiration to accord to victory.
A is true but R is false.
A is false but R is true.
A is true and R is the correct explanation of A.
A is true but R is not the correct explanation of A.

(e) The trending egg lost 150,000 likes and Messi gained 6 million by evening. State any one inference that can be drawn from this. [1]

SECTION- B (Writing Skill And Grammar)

Q.3 Attempt any one of the following: [5]

A. You are P. Lal, Secretary, Science Association of your college. You have read an advertisement in a newspaper by Indian Airlines Corporation offering a reduction in fares for students travelling in a group. Write a letter to this corporation to inquire about the following:

1. Reduction allowed.
2. Whether reduction is applicable to one way or both sides.
3. Whether additional transport facilities offered.
4. The time of take- off and landing.

OR

B. You are Param, office in-charge at Vayusena school, Sant colony, Mumbai. Place an order of stationery items for the school office Modern Store, New Market, Mumbai.

Q.4 Complete the following (Any Eight) as per the instruction [8]

i. Read the given sentence from a recipe review article. Identify the error and supply the correction in the sentence.
This delightful recipe must keep your hunger pangs at bay with it's balanced spices and oriental flavour.
Use the given format for your response.

Error	Correction

ii. Abhiash and Neha had a conversation about the inauguration of Neha's Dance Academy, Report Abhilash's questions.
Is your best friend helping you in this venture?

iii. Fill in the blank by choosing the correct option, to complete the concluding line of an issued circular by an organisation to its Managers.
A copy of the plan is enclosed and _____ (that/than/ this) may be communicated to all Team Leaders for compliance.

iv. Report the dialogue between a vendor and his customer, by completing the sentence:
Vendor : It is nice to see you, Sir!
Customer : Yes, indeed! unlike last month, I have been away for quite some time this month.
The vendor greeted his customer respectfully and mentioned that he was pleased to see him. The customer answered in the affirmative and explained that _____

v. Fill in the blank by choosing the correct option to complete the live feed on a school news channel.
Congratulations Change Makers!
The regional competition awards have been announced and our school music club _____ the first prize.

A. had been won
B. will win
C. won
D. has won

vi. Complete the line from a self- awareness song by filling the blank with the correct option.

The river runs through your veins
The trees and mountains _____ your name
The moon and stars watch over
You're guiding your way back home
To dream the night away

A. will call B. call C. had called D. calls

vii. Read the dialogue between Shabnam and her mentor, Sara regarding her summer internship programme.

Sara : Why did you choose to participate in this internship programme?

Shabnam : Ah! I am convinced this programme has the potential to enhance my abilities.

Sara asked Shabnam _____ in that internship programme. Shabnam sighed and exclaimed that she was convinced that programme had the potential to enhance her abilities.

A. why to choose to participate B. to choose participation
C. why she had chosen to participate D. with her choice in participating

viii. Identify the error; supply correction for the given sentence from a commercial company's current marketing strategy.

The company aimed at increasing authority in areas frequently visited by the clients.

Use the given format for your response.

Error	Correction

ix. Complete the given narrative by filling the blank with the correct option.
The experience of nursing an injured bird left me _____ grateful for knowing the importance of being kind and compassionate to all creatures.

A. feeling B. having felt C. felt D. feels

SECTION- C (LITERATURE)

Q.5 Read the given extracts and answer the questions for Any One of the two given. [3]

A. Ausable moved to an armchair and sat down heavily. "I'm going to raise the devil with the management this time, and you can bet on it," he said grimly this is the second time in a month that somebody has got into my room through that nuisance of a balcony.

i. Ausable's annoyance at the balcony when he says 'nuisance of a balcony'. could be called fake. Why is it so?

ii. Which word replaces 'sat down heavily' in the sentence- 'Ausable moved to an armchair and sat down heavily?

a. dumped b. Humped
c. pumped d. jumped

iii. Delnazz is a new tenant in her locality and Mrs. Mehta is the landlady. Choose the option that lists an action that would require Mrs. Mehta to raise the devil with Delnazz in the similar manner as mentioned in the passage.

a. She uses the elevator five times a day.
b. She feeds the stray dogs in the street.
c. She works late at night and stays home in the afternoon only.

d. She plays music louder than is necessary late into the night.

OR

B. The train pulled out of the station; Pranjol buried his nose in his detective book again. Rajvir too was an ardent fan of detective stories but, at the moment he was keener on looking at the beautiful scenery. It was green, green everywhere. Rajvir had never seen so much greenery before. Then the soft green paddy fields gave way to tea bushes. It was a magnificent view. Against the backdrop of densely wooded hills a sea of tea bushes stretched as far as the eye could see. Dwarfing the tiny tea plants were tall, sturdy, shade- trees and amidst the orderly rows of bushes busily moved doll- like figures.

i. State any one inference about Rajvir from the given context:
"Rajvir too was an ardent fan of detective stories, but at the moment he was keen on looking at the beautiful scenery."

ii. State True or False:
The reason Rajvir was interested in looking at the beautiful scenery was that the detective book he was reading was not interesting.

iii. Which phrase would correctly substitute "doll-like figures" in the given sentence from the extract?

Q.6 Read the given extracts and answer the questions for Any One of the two given. [3]

A. Though to distinguish beasts of prey
A novice might nonplus,
The crocodiles you always may
Tell from the Hyena thus:
Hyenas come with merry smiles;
But if they weep they're crocodiles.

i. Identify the figure of speech in the line "A novice might nonplus." Illustrate.

ii. In this extract the poet has presented two different behavioural traits of animals. They are crocodiles _____, Hyenas _____.

iii. The poet uses 'A novice might nonplus' instead of the novice might get confused. How does this usage impact the poem?

a. adds an incorrect option. b. it is associated with beasts.
c. it rhymes and adds humour. d. it simplifies the meaning.

OR

B. An ultimate shaking grief fixes the boy
As he stands rigid, trembling, staring down
All his young days in the harbour where
His ball went. I would not intrude on him

i. The poet uses the word 'ultimate' to describe the boy's reaction.
Pick the meaning that DOES NOT display what, 'ultimate' means in the context given.

a. Consequent b. Final
c. Conclusive d. Fateful

ii. The boy is very young in this poem. As a mature, balanced grown- up, he might look back and think that his reaction to 'ultimate shaking grief' was -

1. disproportionate to the loss.
2. pretension to procure a new toy.
3. according to his exposure and experience then.
4. a reaction to the failure of retrieving the toy.
5. justified and similar to what it would be currently.

a. 5 & 2 b. 1 & 3 c. 2 & 4 d. 3 & 4

iii. Pick the option that lists the boy's thoughts, matching with the line- As he stands rigid, trembling staring down.

i. I wonder how I
can retrieve the ball.
I guess I'll need help



ii. Good riddance!
Now I can get a new
one I've been
waiting for a while.

iii. I think I can
catch it if I run
through this alley.
I must cut it off
before it falls into
the harbour.

iv. Oh! I've lost it. It's
gone. It's gone
forever. That was all
the time I had with
it. That time is over

a. Option (i) b. Option (ii) c. Option (iii) d. Option (iv)

Q.7 Answer Any Two questions in about 40-50 words: [6]

i. Mijbil turned out to be a playful pet. Elaborate.

ii. Why does the poet hold with those who favour fire?

iii. Justify the opinion that the traditional baker and his bread play a significant role in the culture and social fabric of Goan society?

Q.8 Answer Any One of the following in about 40-50 words: [3]

i. How does Anil's character influence the thief's transformation?

ii. Why does Griffin burn landlord's house and what does this reveal about this personality?

Q.9 Answer Any One of the following in the about 100-120 words: [5]

i. A desire is more personal whereas a vision has broader connotations. You have been asked to present an analysis of Valli's desire and Mandela's vision and how they are respectively fulfilled. Write this presentation draft including your insights, in about 120 words, comparing the approach of both Valli and Mandela
You may begin like this
One acknowledges that both, Valli and Mandela _____, however,

ii. Compare and contrast the experiences of the young seagull in "His First Flight" and the pilot in "The Black Aeroplane." How do both stories reflect the theme of courage and trust?

Q.10 Answer Any One of the following in about 100-120 words. [5]

i. Both Griffin in "Footprints Without Feet" and the thief in "A Question of Trust" possess intelligence, confidence, and resourcefulness. Yet their actions reveal a sharp contrast in values and outcomes. Discuss how the theme of deception and misplaced trust is portrayed through the actions of these two characters. What moral lesson does each story convey through the characters' downfall? Support your answer with textual evidence and critical insight.

ii. Mrs. Pumphrey thought Tricki was suffering from a serious illness, but Herriot knew the real problem. What does this story teach us about overindulgence and responsible pet care?

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Divyansh Singh
X - A
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II-PERIODIC TEST : 2025-26

CLASS – X (CBSE)

Time: 2 hrs. **HINDI COURSE 'A' (002)** M.M.: 50

निर्देश:–1. सभी प्रश्नों के उत्तर निर्देशानुसार देना अनिवार्य है।
2. यथासंभव सभी प्रश्न क्रम से हल करें।

खण्ड–'क' (अपठित बोध)

Q.1. निम्नलिखित अपठित गद्यांश को ध्यानपूर्वक पढ़कर निर्देशानुसार प्रश्नों के उत्तर दें –

अच्छा नागरिक बनने के लिए भारत के प्राचीन विचारकों ने कुछ नियमों का प्रावधान किया है। इन नियमों में वाणी और व्यवहार की शुद्धि, कर्तव्य और अधिकार का समुचित निर्वाह, पारम्परिक सद्भाव और सेवा की भावना आदि हैं। ये सभी नियम यदि एक व्यक्ति के चारित्रिक गुणों के रूप में अनिवार्य माने जाएँ, तो उसका अपना जीवन सुखी और आनंदमय हो सकता है। सभी गुणों का विकास एक बालक में यदि उसकी बाल्यावस्था में ही किया जाए तो वह अपने देश का श्रेष्ठ नागरिक बन सकता है। वाणी व व्यवहार की मधुरता सभी के लिए सुखदायक होती है, समाज में हार्दिक सद्भाव की वृद्धि करती है, किन्तु अहंकारहीन व्यक्ति ही स्निग्ध वाणी और शिष्ट व्यवहार का प्रयोग कर सकता है। अहंकारी और दंभी व्यक्ति सदा अशिष्ट वाणी और व्यवहार का अभ्यासी होता है, जिसका परिणाम यह होता है कि ऐसे आदमी के व्यवहार से समाज में शांति और सौहार्द का वातावरण नहीं बनता।

अ– निम्नलिखित बहुविकल्पीय प्रश्नों के सही उत्तर चयन करके लिखिए :– [1X3=3]

1. अच्छा नागरिक बनने के लिए नियमों का प्रावधान आवश्यक है, क्योंकि यह–
   (क) स्वतंत्रता को बढ़ावा देता है।
   (ख) व्यक्तित्व को निखारकर जीवन को सुखी व आनन्दमय बनाता है।
   (ग) व्यक्तित्व को निखारकर जीवन को मंगलकामना से परिपूर्ण करता है।
   (घ) व्यक्ति को अहंकार, स्निग्ध वाणी और शिष्ट व्यवहार से पूर्ण करता है।
2. वाणी और व्यवहार की मधुरता सभी के लिए सुखदायक होती है। इस कथन के लिए उपयुक्त तर्क है –
   (क) देश के सम्मान, सम्पत्ति और स्वाभिमान को ठेस पहुँचती है।
   (ख) देश व समाज में शांति और सौहार्द का वातावरण नहीं बनता।
   (ग) कर्तव्य और अधिकार का समुचित निर्वाह बहुत आवश्यक है।
   (घ) समाज में हार्दिक सद्भाव की वृद्धि और सुख की प्रतिष्ठा होती है।
3. कथन (A) व्यक्ति को स्निग्ध वाणी और शिष्ट व्यवहार का प्रयोग करना चाहिए।
   कारण (R) व्यक्ति के ऐसे व्यवहार से समाज में शान्ति और सौहार्दपूर्ण वातावरण बनता है
   (क) कथन (A) तथा कारण (R) दोनों गलत हैं।
   (ख) कथन (A) गलत है, किन्तु कारण (R) सही है।
   (ग) कथन (A) सही है, किन्तु कारण (R) कथन (A) की सही व्याख्या नहीं करता।
   (घ) कथन (A) और कारण (R) दोनों सही हैं तथा कारण (R) कथन (A) की सही व्याख्या करता है।

ब– अच्छा नागरिक बनने के लिए किन नियमों का प्रावधान किया गया है ? [2]

Q.2. निम्नलिखित अपठित काव्यांश को ध्यानपूर्वक पढ़कर पूछे गए प्रश्नों के उत्तर निर्देशानुसार दीजिए –

उड़ चल हारिल लिए हाथ में
यही अकेला ओछा तिनका
उषा जाग रही प्राची में
कैसी बाट, भरोसा किनका!
शक्ति रहे तेरे हाथों में
छूट न जाए यह चाह सृजन की
शक्ति रहे तेरे हाथों में
रुक न जाय यह गति जीवन की!
ऊपर ऊपर ऊपर ऊपर
बढ़ा चीर चल दिग्मंडल
अनथक पंखों की चोटों से
नभ में एक मचा दे हलचल

तिनका तेरे हाथों में है
अमर एक रचना का साधन
तिनका तेरे पंजे में है
विधना के प्राणों का स्पंदन!
काँप न यद्यपि दसों दिशा में
तुझे शून्य नभ घेर रहा है
रुक न यद्यपि उपहास जगत का
तुझको पथ से हेर रहा है!
उषा जाग उठी प्राची में
आवाहन यह नूतन दिन का! [1X3=3]

अ– सही विकल्प चुनकर उत्तर दीजिए –

1. उपर्युक्त काव्यांश की किन पंक्तियों में साधन की साधरणता के संकल्प पर विश्वास जताया गया है ?
   (क) उड़ चल हारिल लिए हाथ में, यही अकेला ओछा तिनका।
   (ख) ऊपर ऊपर ऊपर ऊपर, बढ़ा चीर चल दिग्मंडल
   (ग) तिनका पथ की धूल स्वयं तू, है अनंत की पावन धूली।
   (घ) उषा जाग उठी प्राची में, आवाहन यह नूतन दिन का।

2. 'विधाता के प्राणों का स्पंदन!' कवि का आशय है –
   विधाता की ———— शक्ति की झलक।
   (क) रचनात्मक (ख) सृजनात्मक
   (ग) संहारक (घ) उपकारक

3. कथन (A) ऊपर ऊपर ऊपर ऊपर, बढ़ाचीर चल दिग्मंडल।
   कारण (R) अनथक पंखों की चोटों से, नभ में एक मचा दे हलचल।
   (क) कथन (A) गलत है, किन्तु कारण (R) सही है।
   (ख) कथन (A) और कारण (R) दोनों गलत हैं।
   (ग) कथन (A) सही है, और कारण (R) कथन (A) की सही व्याख्या है।
   (घ) कथन (A) सही है, किन्तु कारण (R) कथन (A) की सही व्याख्या नहीं है।

ब– काव्यांश के माध्यम से कवि क्या संदेश देना चाह रहे हैं ? [2]

खण्ड–'ख' (व्यावहारिक व्याकरण)

Q.3. वाक्य पर आधारित निम्नलिखित चार में से किन्हीं तीन प्रश्नों के लघु उत्तर दीजिए :– [1X3=3]
(1) रचना की दृष्टि से वाक्य के भेदों के नाम बताइए।
(2) जो कक्षा में प्रथम आया है, वह मेरा भाई है। (रेखांकित उपवाक्य का उपभेद बताइए)
(3) मूर्तिकार ने सुना और जवाब दिया। (सरल वाक्य में बदलिए)
(4) उसने स्टेडियम जाकर क्रिकेट मैच देखा। (संयुक्त वाक्य में बदलिए)

Q.4. वाच्य पर आधारित निम्नलिखित चार में से किन्हीं तीन प्रश्नों के उत्तर दीजिए :– [1X3=3]
(1) किस वाच्य में केवल सकर्मक क्रिया का प्रयोग होता है ?
(2) बालगोविन भगत कबीर को 'साहब' मानते थे। (कर्मवाच्य में बदलिए)
(3) चिड़ियों से उड़ा नहीं जाता। (वाच्य का भेद बताइए)
(4) नवाब साहब द्वारा संगति के लिए उत्साह नहीं दिखाया गया। (कृतवाच्य में बदलिए)

Q.5. पद–परिचय पर आधारित निम्नलिखित चार में से तीन प्रश्नों के उत्तर दीजिए :– [1X3=3]
(1) पद किसे कहते हैं ?
(2) यह बालक नेधावी है। रेखांकित पद का परिचय दीजिए।
(3) वाह! भारत मैच जीत गया। रेखांकित पद का परिचय दीजिए।
(4) हम कल दिल्ली जा रहे हैं। व्याकरण की दृष्टि से रेखांकित पद क्या है ?

Q.6. अलंकारों पर आधारित निम्नलिखित चार में से तीन प्रश्नों के उत्तर दीजिए :– [1X3=3]
(1) "सिर फट गया उसका वहीं, मानो अरुण रंग का घड़ा।" अलंकार का नाम लिखिए।
(2) जहाँ प्रकृति पर मानवीय चेष्टाओं का आरोप हो, वहाँ कौन सा अलंकार होता है ?
(3) 'देख लो साकेत नगरी है यही, स्वर्ग से मिलने गगन में आ रही।' अलंकार बदाइए।
(4) 'पीपर पात सरिस मन डोला।' अलंकार का नाम लिखिए।

खण्ड–'ग' (पाठ्य पुस्तक एवं पूरक पुस्तक)

Q.7. निम्नलिखित गद्यांश को पढ़कर पूछे गए बहुविकल्पीय प्रश्नों के उत्तर दीजिए – [1X4=4]

कार्तिक आया नहीं कि बालगोबिन भगत की प्रभातियाँ शुरू हुईं, जो फागुन तक चला करतीं। इन दिनों वह सबेरे ही उठते। न जाने किस वक्त जगकर वही नदी–स्नान को जाते–गाँव से दो मील दूर। वहाँ से नहा–धोकर लौटते और गाँव के बाहर ही, पोखरे के ऊँचे भिंडे पर, अपनी खँजड़ी लेकर जा बैठते और अपने गाने टेरने लगते। मैं शुरू से ही देर तक सोने वाला हूँ, किन्तु एक दिन, माघ की उस दाँत किट किटाने वाली भोर में भी, उनका संगीत मुझे पोखरे पर ले गया था। अभी आसमान के तारों के दीपक बुझे नहीं थे। हाँ, पूरब में लोही लग गई थी। जिसकी लालिमा को शुक्र तारा और बढ़ा रहा था। खेत, बगीचा, घर सब पर कुहासा छा रहा था। सारा वातावरण अजीब रहस्य से आवृत्त मालूम पड़ता था। उस रहस्यमय वातावरण में एक कुश की चटाई पर पूरब मुँह, काली कमली ओढ़े भगत अपनी खँजड़ी लिए बैठे थे। उनके मुँह से शब्दों का ताँता लगा था, उनकी अँगुलियाँ खँजड़ी पर लगातार चल रही थीं।

1. प्रभातियाँ किस समय गाया जाने वाला गीत है ?
   (क) भोर (ख) संध्या
   (ग) दोपहर (घ) रात
2. 'उनके मुँह से शब्दों का ताँता लगा था।' – का आशय है–
   (क) वह एक के बाद एक गीत गाए जा रहे थे।
   (ख) वह लगातार प्रवचन दिए जा रहे थे।
   (ग) वह लगातार लोगों को आशीर्वचन दे रहे थे।
   (घ) वह लगातार किसी से बातें किए जा रहे थे।
3. गद्यांश में किस ऋतु का उल्लेख है ?
   (क) वसंत ऋतु का (ख) ग्रीष्म ऋतु का
   (ग) शीत ऋतु का (घ) वर्षा ऋतु का
4. बालगोबिन भगत की प्रभातियाँ कब से कब तक चला करती थीं ?
   (क) फागुन से कार्तिक तक (ख) कार्तिक से फागुन तक
   (ग) फागुन से आषाढ़ तक (घ) कार्तिक से चैत्र तक

Q.8. गद्य–पाठों पर आधारित निम्नलिखित तीन प्रश्नों में से किन्हीं दो प्रश्नों के उत्तर दीजिए– [2X2=4]

(क) जब तक हालदार साहब ने कैप्टन को साक्षात् नहीं देखा था, तब तक उनके मानस पटल पर उसका कौन–सा चित्र रहा होगा ?

(ख) लेखक ने क्या सोचकर सेकंड क्लास का टिकट लिया ?

(ग) वास्तविक अर्थों में 'संस्कृत व्यक्ति' किसे कहा जा सकता है ?

Q.9. निम्नलिखित पद्यांश पर आधारित प्रश्नों के सही विकल्प चुनकर उत्तर लिखिए – [1X4=4]

नाथ संभुधनु भंजनिहारा। होइहि केउ एक दास तुम्हारा।।
आयेसु काह कहिअ किन मोहि। सुनि रिसाइ बोले मुनि कोही।।
सेवकु सो जो करै सेवकाई। अरि करनी करि करिअ लराई।।
सुनहु राम जेहि सिवधनु तोरा। सहसबाहु सम सो रिपु मोरा।।

1. किसका धनुष तोड़ दिया गया था ?
   (क) राम का (ख) दशरथ का
   (ग) शिवाजी का (घ) परशुराम का
2. प्रथम पंक्ति किसने कही है ?
   (क) राम ने (ख) लक्ष्मण ने
   (ग) परशुराम ने (घ) विश्वामित्र ने
3. 'सुनि रिसाइ बोले मुनि कोही' में मुनि कौन हैं ?
   (क) विश्वामित्र (ख) परशुराम
   (ग) वशिष्ठ (घ) दुर्वासा
4. 'सहसबाहु सम सो रिपु मोरा' में अलंकार बताएं–
   (क) उपमा (ख) अतिशयोक्ति
   (ग) यमक (घ) उत्प्रेक्षा

Q.10. काव्य–खण्ड के पाठों पर आधारित निम्नलिखित तीन प्रश्नों में से किन्हीं दो प्रश्नों के उत्तर दीजिए– [2X2=4]

(क) कृष्ण के प्रति अपने अनन्य प्रेम को गोपियों ने किस प्रकार अभिव्यक्त किया ?

(ख) लक्ष्मण ने वीर योद्धा की क्या–क्या विशेषताएँ बताई ?

(ग) 'उज्ज्वल गाथा कैसे गाऊँ, मधुर चाँदनी रातों की'– कथन के माध्यम से कवि क्या कहना चाहता है ?

Q.11. कृतिका के पाठों पर आधारित निम्नलिखित दो प्रश्नों में से किसी एक प्रश्न का उत्तर दीजिए। [4X1=4]

(क) लेखक ने अपने आपको हिरोशिमा के विस्फोट का भोक्ता कब और किस तरह महसूस किया ?

(ख) पाठ–माता का आँचल में माता–पिता का बच्चे के प्रति जो वात्सल्य व्यक्त हुआ है, उसे अपने शब्दों में लिखिए।

Q.12. निम्नलिखित में से किसी एक विषय पर दिए गए संकेत–बिन्दुओं के आधार पर 80–100 शब्दों का एक अनुच्छेद लिखिए – [4X1=4]

(क) संयुक्त परिवार – आज की आवश्यकता
- (1) युवा पीढ़ी को रिश्तों का ज्ञान
- (2) मिल–जुलकर रहने की भावना
- (3) अच्छे संस्कारों का ग्रहण
- (4) बुजुर्गों की उपस्थिति व महत्ता
- (5) अकेलेपन से मुक्ति

(ख) इंटरनेट – एक संचार क्रांति
- (1) आधुनिक युग विज्ञान का युग
- (2) इंटरनेट एक विशेष उपलब्धि
- (3) इंटरनेट के लाभ
- (4) इंटरनेट से संचार जगत में क्रांति

(ग) पराधीन सपनेहुँ सुख नाहीं
- (1) पराधीनता का अर्थ
- (2) स्वतंत्रता का महत्व
- (3) पराधीनता कलंक
- (4) पराधीनता से मुक्ति के उपाय

Q.13. आपके मोहल्ले में एक नया पब्लिक स्कूल खुला है। उसके लिए लगभग 40 शब्दों में विज्ञापन तैयार कीजिए। [4X1=4]

अथवा

आप मोहन/मोहिनी है आपके मित्र ने नीट की परीक्षा में राज्य में प्रथम स्थान प्राप्त किया है। उसे बधाई देते हुए लगभग 40 शब्दों में एक संदेश लिखिए।

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Divyansh Singh
X-A
(21)

LPSC 25

II PERIODIC TEST : 2025-26
CLASS – X (CBSE)
MATHEMATICS

Maximum Marks : 50
Time allowed : 2 hours

General Instructions:
(i) All question are compulsory.
(ii) This question paper comprises of four sections A, B, C and D.
(iii) Section – A, contains questions number 1 to 5 of one mark each.
(iv) Section – B, contains questions number 6 to 10 of two marks each.
(v) Section – C, contains questions number 11 to 15 of three marks each.
(vi) Section – D, contains questions number 16 to 20 of four marks each.
(vii) Internal choice is also given in some questions.

SECTION – 'A' [1×5=5]

Q.1. The quadratic polynomial whose zeroes are – 3 and 4 is
(a) $x^2 - x + 12$ (b) $x^2 + x + 12$
(c) $x^2 + x - 12$ (d) $x^2 - x - 12$

Q.2. If $\sin A = \frac{15}{17}$, then cot A will be :
(a) $\frac{15}{8}$ (b) $\frac{8}{15}$ (c) $\frac{8}{17}$ (d) $\frac{17}{15}$

Q.3. If the length of the shadow of a tower is equal to its height, then the angle of elevation of the sun is :
(a) 45° (b) 30° (c) 60° (d) 90°

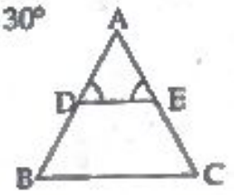
Q.4. The area of a circle is 49 π cm². Its circumference is :
(a) 7 π cm (b) 14 π cm
(c) 21 π cm (d) 28 π cm

Q.5. If a pair of linear equations is inconsistent then their graph lines will be :
(a) parallel (b) always coincident
(c) always intersecting (d) intersecting or coincident

SECTION – 'B' [2×5=10]

Q.6. Evaluate : $\tan 30° . \operatorname{cosec} 60° + \tan 60° . \sec 30°$

Q.7. If D and E are points on the sides AB and AC respectively of ΔABC such that AB = 5.6 cm, AD = 1.4 cm, AC = 7.2 cm and AE = 1.8 cm, show that DE||BC.



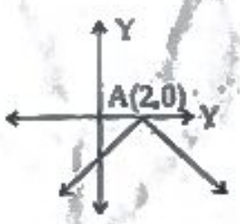
Q.8. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 10 minutes.

Q.9. A game of chance consists of spinning an arrow which comes to rest pointing one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 and these ae equally likely outcomes. What is the probability that it will point at (a) an odd number (b) perfect square number

Q.10. If the sum of zeroes is equal to product of zeroes in the polynomial, $p(x) = ax^2 + bx + c$, then find the value of b + c.

OR

The graph of polynomial $p(x) = kx^2 + 4x - 4$ is given, find the value of k.



SECTION – 'C' [3×5=15]

Q.11. Prove that $\sqrt{5}$ is an irrational number.

Q.12. D is a point on the side BC of a triangle ABC such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB.CD$.

OR

E is a point on the side AD produced of a ||gm ABCD and BE intersects CD at F. Show that :

(i) $\Delta ABE \sim \Delta CFB$ (ii) $\frac{AB}{CF} = \frac{BE}{FB}$

Q.13. Find the coordinates of the points which divide the line segment joining A(–2, 2) and B(2, 8) into four equal parts.

Q.14. A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle in to 10 equal sectors. Find:

(i) The total length of the silver wire required.

(ii) The area of each sector of the brooch.

Q.15. If A and B are acute angles such that $\sin(A - B) = 0$ and $2\cos(A + B) = 1$, then find the angles A and B.

SECTION – 'D' [4×5=20]

Q.16. A man standing on the deck of a ship, which is 10 m above water level. He observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of the hill as 30°. Calculate the distance of the hill from the ship and the height of the hill.

Q.17. The age wise participation of students in the Annual Function of a school is shown in the following distribution :

Age in years	5–7	7–9	9–11	11–13	13–15	15–17	17–19
No. of students	x	15	18	30	50	48	x

Find the missing frequencies (x), when the sum of frequencies is 181. Also find the mode of the data.

Q.18. Prove that : $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$

Q.19. State and prove Basic Proportionality Theorem :

OR

The diagonals of quadrilateral ABCD intersect each other at the point O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show tha ABCD is frapezium.

CASE STUDY QUES.

Q.20. Essel World is one of India's largest amusement parks. The ticket charges for the park are ₹ 150 per child and ₹ 250 per adult .
On a day, the cashier of the park found that 300 tickets were sold and an amount of ₹ 55,000 was collected.
Based on the above information, answer the following questions.

(i) If the number of children visited is x and the numbers of adults visited is y then write the given situation algebraically.

(ii) (a) How many children visited the amusement park that day?

OR

(b) How many adults visited the amusement park that day.

(iii) How much amount will be collected if 250 children and 100 adults visit the amusement park?

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